

# The Influence of Citation Formatting on Human Trust in Fabricated References

*A Review of Evidence on Citation Presentation, Cognitive Heuristics, and the Credibility of Non-Existent Sources in the Generative-AI Era*

## Abstract

The mass production of scholarly-looking text by large language models (LLMs) has made a specific failure mode newly consequential: the fabricated citation, an invented reference that is bibliographically well-formed but points to no real work. This paper reviews the interdisciplinary evidence on why fabricated references are frequently believed rather than questioned, situating the problem at the intersection of citation-fabrication research, cognitive psychology, and credibility science. Empirical studies show that generative chatbots invent between roughly one-fifth and over half of the references they produce depending on model and domain (Walters & Wilder, 2023; Cabezas-Clavijo & Sidorenko-Bautista, 2025; Alkaissi & McFarlane, 2023), and that these fabrications have already produced real-world sanctions in legal practice (Weiser, as documented in Magesh-adjacent scholarship on legal hallucination detection; Liu et al., 2026). Drawing on the heuristic-systematic model of persuasion, the illusory-truth and cognitive-fluency literatures, and the "seductive allure" effect documented for scientific-looking visual and numeric detail, the paper argues that formal citation apparatus functions as a peripheral authority cue that is processed heuristically rather than verified. It further reviews new experimental evidence that this susceptibility is not confined to humans: large language models themselves are more likely to accept false claims when a fabricated citation is attached (Khurana et al., 2026). The paper synthesizes current detection and verification approaches, identifies methodological and educational gaps, and proposes directions for research on format-sensitive trust calibration.

**Keywords:** citation fabrication; source credibility; cognitive heuristics; large language model hallucination; academic integrity; misinformation; illusory truth effect; reference verification

## 1. Introduction

A citation is, in ordinary scholarly practice, a promissory note: it tells the reader that a claim can be checked against an identifiable prior source. The formatting conventions that surround citations — an author-year parenthetical, a numbered superscript, a full reference entry with a journal name, volume, and digital object identifier (DOI) — exist to make that promise verifiable. Yet a growing body of evidence shows that the presentational apparatus of a citation is frequently treated by readers, and even by other artificial intelligence systems, as sufficient evidence of a claim's truth, independent of whether the underlying source exists at all (Khurana et al., 2026). This decoupling of citation form from citation substance has become acute since 2022, as generative language models capable of producing fluent, correctly formatted, and entirely invented references have moved from a laboratory curiosity to a default drafting tool

used by students, clinicians, journalists, and even attorneys (Alkaissi & McFarlane, 2023; Walters & Wilder, 2023).

The problem is not merely that language models occasionally hallucinate. It is that the hallucinations they produce are formatted exactly like real scholarship — with plausible author names, credible-sounding journal titles, believable years, and, not infrequently, invented DOIs or PubMed identifiers — and that this formatting appears to do real persuasive work on human readers (Eika & Sandnes, 2022; StrongMocha, 2026). A 2023 legal case in the Southern District of New York, in which attorneys submitted a brief containing six entirely fictitious judicial opinions generated by ChatGPT, complete with fabricated quotations, is the most widely cited illustration of the phenomenon (documented in detail by researchers studying AI-hallucinated citations in legal contexts; see Section 4). By 2026, benchmarking work had identified over a thousand court filings containing fabricated citations, with the count still rising despite growing judicial sanctions (Liu, Stambach, & Henderson, 2026).

This paper asks a narrower and more tractable question than "why do people believe misinformation": specifically, what is it about the formatting and presentation of citations — as distinct from their content — that shapes human trust, and why does that formatting so often fail to discriminate between real and fabricated sources? To answer this, the paper draws together three normally separate literatures: (1) empirical measurements of citation fabrication rates in generative AI output; (2) cognitive-psychological research on fluency, authority heuristics, and the persuasive power of scientific-looking detail; and (3) emerging computational work on citation verification and on citation-induced bias in both human and machine judgment. The aim is not to catalogue every instance of AI citation failure, but to explain, using converging evidence, why formatted references exert disproportionate influence on trust relative to the verification effort readers actually apply to them.

## 2. Background and Related Work

### *2.1 The Scale and Nature of Citation Fabrication*

The first systematic documentation of AI citation fabrication came from case-based reports in the medical literature. Alkaissi and McFarlane (2023) tested ChatGPT on two clinical case topics and found that the references it generated, while formatted like ordinary journal citations, corresponded either to nonexistent articles or to real articles that did not support the claims attributed to them. Walters and Wilder (2023) then conducted the first large-scale, multi-topic audit: prompting ChatGPT-3.5 and ChatGPT-4 to write short literature reviews across 42 multidisciplinary topics and manually verifying all 636 resulting references against library databases. They found that 55% of GPT-3.5's citations and 18% of GPT-4's citations were entirely fabricated, and that even the non-fabricated citations frequently contained substantive errors in author names, years, volumes, or page numbers — while still nominally adhering to APA formatting conventions (Walters & Wilder, 2023).

Subsequent audits have shown the problem generalizing across models and worsening as scrutiny broadens. Cabezas-Clavijo and Sidorenko-Bautista (2025) evaluated eight consumer chatbots — ChatGPT, Claude, Copilot, DeepSeek, Gemini, Grok, Le Chat, and Perplexity — on 400 bibliographic references spanning five academic domains, and found that only 26.5% of references were fully correct, while 39.8% were either erroneous or entirely invented; two models (Grok and DeepSeek) avoided outright fabrication, while Copilot, Perplexity, and Claude showed the highest hallucination rates in that particular test. Their results also revealed that fabrication rates were not uniform across document types, with journal-article citations more error-prone than book citations, and that different models' outputs overlapped substantially in the specific (often fictitious) sources they generated, suggesting shared statistical patterns rather than independent errors (Cabezas-Clavijo & Sidorenko-Bautista, 2025).

Critically for the present argument, none of this fabrication is haphazard in appearance. Fabricated references are stylistically indistinguishable from real ones at the level of format: they use correct citation punctuation, plausible journal names, and internally consistent author-initial conventions. This is a direct consequence of how large language models generate text — by predicting statistically likely continuations rather than retrieving verified records — so that the model reproduces the surface conventions of scholarly citation with high fidelity even while inventing the substance underneath (Alkaissi & McFarlane, 2023). The result is a category of misinformation that is uniquely difficult to detect through format inspection alone, because the format is, by construction, correct.

## *2.2 Citation Fabrication Beyond Academic Writing*

The consequences of citation fabrication are not confined to scholarship. In law, Liu, Stambach, and Henderson (2026) document over a thousand identified court filings containing AI-fabricated citations, a number that has grown year over year despite sanctions, and argue that the adversarial structure of litigation — in which opposing counsel is strongly incentivized to check every cited case — is precisely what exposes these fabrications, a check that has no equivalent in ordinary reading or even in much of peer review (Liu et al., 2026). This observation matters for the present paper because it implies that citation-formatting trust is not primarily corrected by readers' own scrutiny; it is corrected, when it is corrected at all, by adversarial or institutional incentives external to the reading act itself.

In healthcare communication, fabricated citations that reach patient-facing forums, wellness blogs, or advocacy sites are especially concerning because the audience typically lacks the domain expertise or database access needed to verify a reference, and because a citation's mere presence has been shown to increase message credibility for lay readers regardless of whether the source is checked (Eika & Sandnes, 2022; StrongMocha, 2026).

## *2.3 Citation Presence as a Credibility Signal: Prior Human-Subjects Research*

Independent of the AI-fabrication literature, communication and information-science researchers have long studied how citation-related cues shape perceived credibility. Metzger and Flanagin (2013) synthesized evidence that online readers rely on a small set of cognitive heuristics — including reputation, endorsement, consistency, and persuasive-intent heuristics — to make rapid credibility judgments that substitute for effortful verification; the presence of a citation or reference list functions as a reputation cue within this framework, signaling that a claim has already been vetted elsewhere. Eika and Sandnes (2022), studying how students judge the trustworthiness of publications based on visible reference-list attributes, found that undergraduate and graduate readers were most strongly swayed by citation counts and publication recency, while author identity and publication venue exerted comparatively less influence — evidence that readers respond to bibliometric-looking surface signals even when they lack the expertise to evaluate the underlying scholarship.

A parallel qualitative literature on health-information credibility corroborates this pattern outside the academic context: parents evaluating online health news for their adolescents reported using reputation, endorsement, and consistency heuristics, and explicitly cited the presence of numbers, statistics, and linked scientific research as boosting an article's perceived trustworthiness, largely independent of whether they followed those links (qualitative heuristics research on parental credibility judgments; see Section 3). Together, this body of work establishes that citation-like apparatus operates as what persuasion researchers call a peripheral cue: a feature of message presentation that shapes acceptance through heuristic rather than deliberative processing (Chaiken, 1980, as applied in later heuristic-systematic model research).

### **3. Main Discussion: Why Citation Formatting Shapes Trust**

#### ***3.1 Cognitive Fluency and the Illusory Truth of Formatted Claims***

The most parsimonious psychological explanation for why fabricated but well-formatted citations are believed is cognitive fluency: the subjective ease with which a stimulus is processed. Fluently processed information is judged, all else equal, as more likely to be true, a bias that operates independently of whether the reader consciously endorses the underlying claim (cognitive fluency and judgments of truth research, synthesized in dual-process terms). This fluency-truth link underlies the well-established illusory truth effect, in which mere repeated exposure to a claim increases its perceived veracity even without new supporting evidence, with the largest gain in perceived truth occurring after only a second exposure. A correctly formatted citation plausibly recruits this same mechanism: a reference that looks like thousands of other references a reader has previously encountered is processed more fluently, and that fluency is misattributed to the truth of the underlying claim rather than to the familiarity of the format.

This fluency account is reinforced by findings that the fluency-trust link is strongest precisely when readers are anxious, time-pressured, or otherwise motivated to rely on heuristic rather than analytic processing, and is dampened mainly by active "persuasion knowledge" — an explicit

awareness that one is the target of a formatting or rhetorical tactic (cognitive fluency as a driver of trust research). Because most readers of AI-assisted writing are not primed to treat a citation as a persuasion tactic — citations are, after all, conventionally a transparency device rather than a rhetorical one — the ordinary skepticism that might otherwise be applied to marketing claims is not engaged.

### *3.2 The "Seductive Allure" of Scientific-Looking Detail*

A closely related and well-replicated line of research demonstrates that logically irrelevant scientific-looking detail inflates perceived credibility even when it adds no evidential value. In the foundational study, McCabe and Castel (2008) found that adding a brain image to a written argument about cognitive neuroscience increased readers' ratings of the argument's scientific soundness relative to the same argument accompanied by a bar graph, a topographic activation map, or no image at all — despite the image conveying no additional information relevant to the argument's validity. Subsequent replication attempts have complicated the picture: a large multi-site replication by Michael, Newman, Vuorre, Cumming, and Garry (2013), comprising ten experiments and nearly 2,000 participants, found that the original brain-image effect was much smaller than first reported and, in their pooled meta-analysis, close to negligible, prompting caution about the size (though not necessarily the direction) of the effect. Other researchers have since proposed that the persuasive power lies less in the image per se than in the general phenomenon of logically irrelevant, ostensibly technical detail lending unearned authority to a claim — the same underlying mechanism proposed to explain why numeric, quantitative, or bibliographic apparatus can inflate perceived rigor independent of content (research on the reductive allure of scientific explanations).

Extended to citation formatting, this literature supports a specific hypothesis: a fabricated reference bearing a plausible DOI, page range, and journal name may function analogously to McCabe and Castel's brain image — a piece of ostensibly technical, quantifiable detail that readers interpret as evidence of underlying rigor without evaluating whether the detail itself is accurate or even real. This would explain why fabrication rates as high as 40–55% in some studies (Cabezas-Clavijo & Sidorenko-Bautista, 2025; Walters & Wilder, 2023) coexist with widespread, largely unchecked circulation of AI-assisted text: the formatting itself discourages the very scrutiny that would reveal the fabrication.

### *3.3 Authority Heuristics and the Heuristic-Systematic Model*

Persuasion research offers a complementary framework in the heuristic-systematic model, which distinguishes fast, cue-based heuristic processing from slower, effortful systematic processing, and predicts that source-credibility cues function as heuristic shortcuts particularly when a message recipient's motivation or capacity for systematic evaluation is low. Empirical work in this tradition treats explicit citations and quotations from outside experts as textbook credibility signals: news articles containing explicit citations are rated as more credible by readers than otherwise identical articles without them, and this effect appears to generalize from scientific

writing to journalism, marketing, and social media content (Srba et al., 2024). Crucially, this literature has generally measured the effect of citation presence versus absence, not citation authenticity versus fabrication — a distinction that recent AI-era research has begun to interrogate directly (see Section 3.5).

### *3.4 Source Discreditation, Badges, and the Limits of Credibility Signaling*

If citation-like formatting functions as an authority heuristic, an intuitive countermeasure is to attach explicit credibility signals — source ratings, fact-check badges, or discreditation warnings — that readers can use instead of, or alongside, the citation itself. Evidence on this class of intervention is mixed. In simulated social-media environments, dynamically updated source-credibility scores have been shown to improve participants' ability to discriminate true from false posts, and combined credibility-badge and social-norm interventions produced measurable reductions in engagement with misinformation (source-credibility and social-norm intervention research, PMC10960008). Complementary work on source discreditation found that directly attacking a source's trustworthiness — for example, by disclosing a conflict of interest — reduced belief in that source's claims nearly as effectively as directly correcting the false claim itself, with the strongest effect obtained by combining discreditation with correction (misinformation-reduction research on source discreditation).

However, other studies find that source or media-outlet credibility exerts little independent influence on belief once message content is held constant, and a large field experiment testing credibility badges on search-engine results found no measurable effect on belief in either true or false claims from low-quality sources (Aslett et al., 2023, discussing prior null findings on credibility badges). Perhaps most counterintuitively, Aslett, Sanderson, Godel, Persily, Nagler, and Tucker (2023) found across five experiments that searching online specifically to verify the truthfulness of a false news article actually increased belief in that article, an effect concentrated among users whose searches returned corroborating evidence from low-quality sources — a phenomenon the authors term a "data void." Applied to citation verification, this raises an uncomfortable possibility: a reader who attempts to check a fabricated citation by searching for it, rather than consulting an authoritative bibliographic database, may encounter secondary sources that have already propagated the same fabrication, reinforcing rather than correcting the false belief (Aslett et al., 2023; Enago, 2026, on the citation feedback loop).

### *3.5 Citation Bias in Machine Judgment*

A striking recent development is that the authority heuristic associated with citation formatting appears to be reproduced inside large language models themselves, not merely in the human readers who consume their output. Khurana, Ramana RN, and Kumar (2026) introduced AuthorityBench, a 220,564-prompt, four-domain benchmark that independently manipulates claim veracity and citation veracity in a fully crossed factorial design. Across seven evaluated models, they found that the mere presence of a citation — whether real or fabricated — increased hallucination rates relative to a no-citation baseline, and that the effect was most severe

when a fabricated citation was attached to an otherwise true claim: in that condition, models were more likely to wrongly deny a correct fact than in any other condition, with hallucination rates reaching 35–77% in the general-knowledge domain and increases of 3 to over 22 percentage points over baseline (Khurana et al., 2026). Notably, legal-domain claims were comparatively resistant to this effect, and venue prestige (the apparent status of the citing journal) had negligible impact — suggesting that models, like the human readers described in Section 3.3, respond to citation presence as an authority cue rather than genuinely evaluating source quality. The authors frame this as a failure of epistemic reasoning under authority pressure rather than a simple factual-knowledge gap (Khurana et al., 2026), a framing that closely parallels the heuristic-processing account of human citation trust developed above.

This convergence is theoretically significant: it suggests that citation-format-driven trust is not a purely human cognitive quirk attributable to time pressure or motivational limits, but may instead reflect a more general property of systems — biological or artificial — trained or socialized on corpora in which citation presence reliably correlated with claim reliability. If so, the same formatting cues that mislead human readers may also compromise retrieval-augmented generation (RAG) systems that are increasingly deployed specifically to reduce hallucination, since injecting citations into a model's context window can, perversely, increase acceptance of unsupported claims (Khurana et al., 2026).

## 4. Current Approaches and Developments

Responses to citation fabrication have developed along three main tracks: automated verification tooling, editorial and disciplinary policy, and model-level mitigation.

### *4.1 Automated and Semi-Automated Verification*

The dominant technical response has been the development of citation-checking pipelines that cross-reference a candidate reference against authoritative bibliographic databases such as CrossRef, PubMed, Semantic Scholar, and OpenAlex, flagging entries whose DOI does not resolve, whose author-title combination cannot be matched, or whose metadata is internally inconsistent. Commercial and open-source tools of this kind report high detection accuracy against known fabrication patterns, though developers of these systems are careful to note that automated matching can confirm a source's existence but cannot, by itself, judge whether the source actually supports the claim attributed to it — a harder problem sometimes called attribution or evidence verification, as distinct from existence verification (a distinction drawn explicitly in recent reference-hallucination detection research). More sophisticated multi-agent pipelines attempt to close this gap by decomposing verification into claim extraction, evidence retrieval, passage matching, and calibrated judgment stages, explicitly targeting cases where a citation is real but does not support the claim made in its name.

### *4.2 Editorial, Institutional, and Legal Responses*

Academic publishers and preprint servers have begun imposing direct consequences for citation fabrication; arXiv, for example, has instituted bans on authors who submit papers containing fabricated references, and major journals now explicitly hold authors accountable for reference accuracy regardless of whether AI tools were used in drafting (Enago, 2026). In legal practice, courts have sanctioned attorneys in a growing number of jurisdictions since the first widely publicized 2023 case, with financial penalties increasing over time, and benchmarking research has begun evaluating whether AI-based citation-checking agents can be deployed by courts or law clerks to screen filings before submission (Liu et al., 2026). These institutional responses share a common logic: because citation formatting itself provides no reliable signal of authenticity, verification must be relocated to an external, adversarial, or automated check rather than relying on the citing document's own presentation.

### ***4.3 Model-Level and Interface-Level Mitigation***

A third track targets the generative process itself, through retrieval-augmented generation that grounds citations in retrieved documents rather than free generation, through reduced hallucination rates in progressively updated model versions (Walters and Wilder's comparison of GPT-3.5 and GPT-4 found substantial, though incomplete, improvement across a single model generation), and through interface designs that surface source credibility or verification status directly alongside AI-generated citations rather than leaving verification entirely to the reader (video-platform credibility-signal research; Aslett et al.'s social-norm and credibility-badge interventions). However, as discussed in Section 3.5, the AuthorityBench findings complicate the retrieval-augmented-generation solution specifically, since citation injection can increase rather than decrease acceptance of false claims when the injected citation is itself fabricated (Khurana et al., 2026).

## **5. Research Challenges and Limitations**

Several challenges limit both the current evidence base and the effectiveness of proposed mitigations.

## **6. Research Gaps and Future Directions**

Building on the challenges above, several concrete research gaps emerge.

## **7. Conclusion**

Fabricated citations are not merely a technical byproduct of imperfect language models; they are a socio-cognitive problem sustained by the mismatch between how much verification effort a formatted citation appears to warrant and how little it actually receives. The evidence reviewed here converges on a consistent account: citation apparatus — reference lists, DOIs, journal names, numbered or parenthetical in-text markers — functions largely as a peripheral authority cue, processed fluently and heuristically rather than verified systematically, in a manner closely

related to the well-documented seductive allure of other scientific-looking but logically irrelevant detail (McCabe & Castel, 2008; Michael et al., 2013). Empirical audits show that generative AI systems exploit this gap at scale, fabricating a substantial share of the references they produce while preserving perfect surface formatting (Walters & Wilder, 2023; Cabezas-Clavijo & Sidorenko-Bautista, 2025), with consequences that have already reached courtrooms and clinical settings (Liu et al., 2026). Perhaps most striking is the recent demonstration that this same authority heuristic operates inside the language models themselves, such that citation presence — real or fabricated — measurably distorts model judgment (Khurana et al., 2026), suggesting the phenomenon reflects a general property of statistical pattern-based reasoning rather than a purely human failing. Addressing the problem will require moving verification out of the reading act itself, where fluency and authority heuristics dominate, and into dedicated, database-grounded checking infrastructure, institutional accountability, and — based on the gaps identified above — a still largely unbuilt body of controlled research on exactly how citation format, independent of content, shapes what readers and machines alike are willing to believe.

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